

GGL90 Validation Report

MITgcm vs Python Port Comparison (single-step replay)

Generated: 2026-09-27 05:57:01

MITgcm Reference:

File: mitgcm_ggl90_outputs_global_ocean_cs32x15_idemix_10.nc
UUID: 79a90418-e730-49e3-ac21-0cdae8b3015a

Python Port Output:

File: python_ggl90_outputs_global_ocean_cs32x15_idemix_10.nc
UUID: 79a90418-e730-49e3-ac21-0cdae8b3015a

Provenance: ☐ UUIDs match

Timesteps: 10

Grid: 384 × 16 × 15

Fields compared: visc_az, diff_kz, mixing_length, tke_after
(GGL90 single-step replay: Python fed MITgcm's own captured
tke_before(t); only the one-step update is compared, isolating
agreement on the physics formula from trajectory drift.)

GGL90 Field Statistics

Wet-cell comparison (NaN-excluded below-seafloor/above-ice-shelf cells)

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visc_az  (Vertical eddy viscosity (K_m), m^2/s)
          Status:  $\Delta$  CHECK
Median |diff|: 3.9072e-04    P95 |diff|: 4.9916e+00    Max |diff|: 1.0000e+02
Median rel:   3.6737e+04    P95 rel:   5.1923e+05    Max rel:   1.7473e+06
          Fraction >1% rel error: 62.5125%
          Points compared: 813820 (excluded 107780 NaN cells)

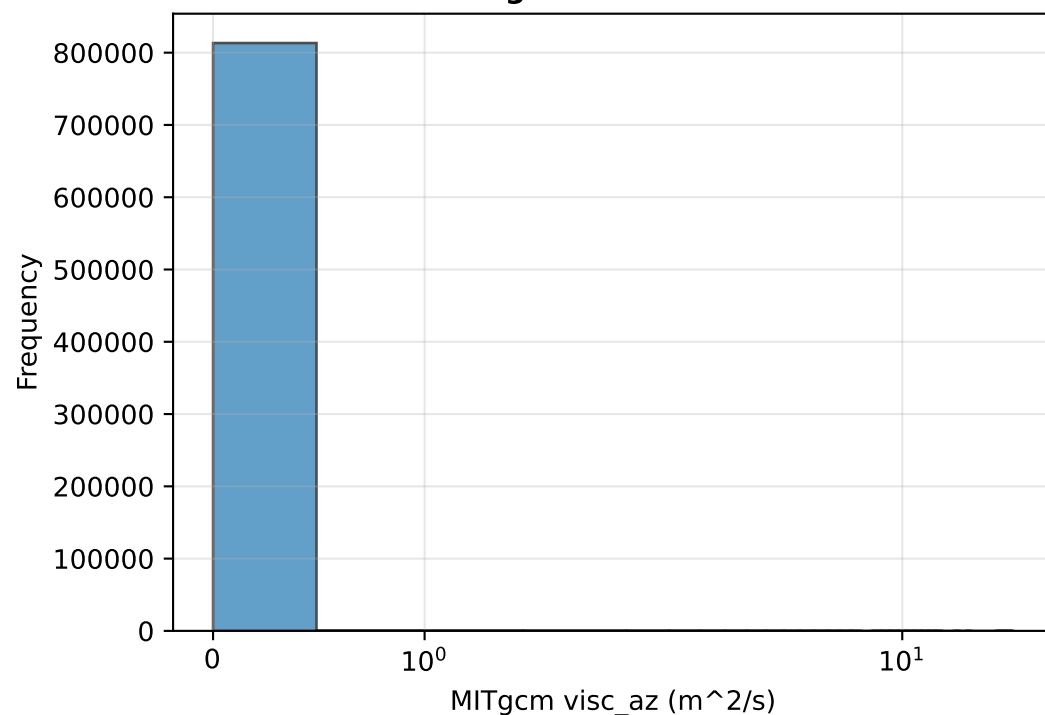
diff_kz  (Vertical eddy diffusivity (K_h), m^2/s)
          Status:  $\Delta$  CHECK
Median |diff|: 7.4892e-01    P95 |diff|: 3.3267e+03    Max |diff|: 2.0149e+09
Median rel:   9.9695e-01    P95 rel:   9.9975e-01    Max rel:   1.0000e+00
          Fraction >1% rel error: 62.7333%
          Points compared: 813820 (excluded 107780 NaN cells)

mixing_length (GGL90 mixing length (L), m)
          Status:  $\Delta$  CHECK
Median |diff|: 5.1744e+02    P95 |diff|: 6.7815e+04    Max |diff|: 1.4492e+07
Median rel:   5.5952e+04    P95 rel:   5.5528e+05    Max rel:   1.7473e+06
          Fraction >1% rel error: 62.7157%
          Points compared: 813820 (excluded 107780 NaN cells)

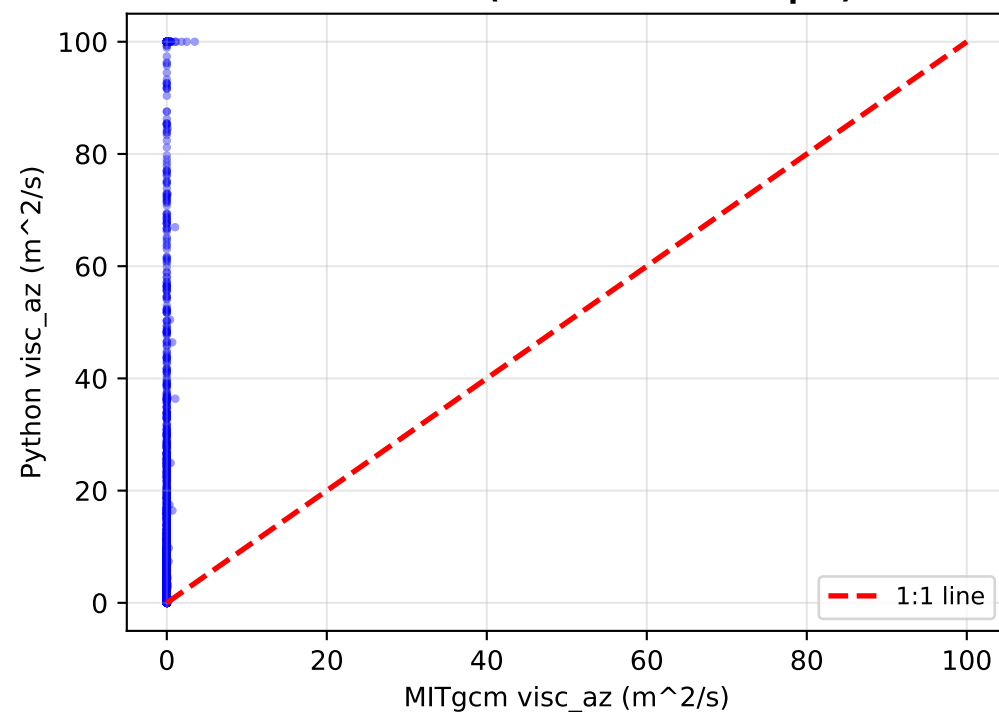
tke_after (TKE at t+1 (single-step replay), m^2/s^2)
          Status:  $\Delta$  CHECK
Median |diff|: 5.1499e-08    P95 |diff|: 5.3796e-02    Max |diff|: 4.1995e+05
Median rel:   3.0171e+01    P95 rel:   2.0207e+06    Max rel:   4.3265e+15
          Fraction >1% rel error: 64.7700%
          Points compared: 813820 (excluded 107780 NaN cells)
```

visc_az (Vertical eddy viscosity (K_m)) -- MITgcm vs Python

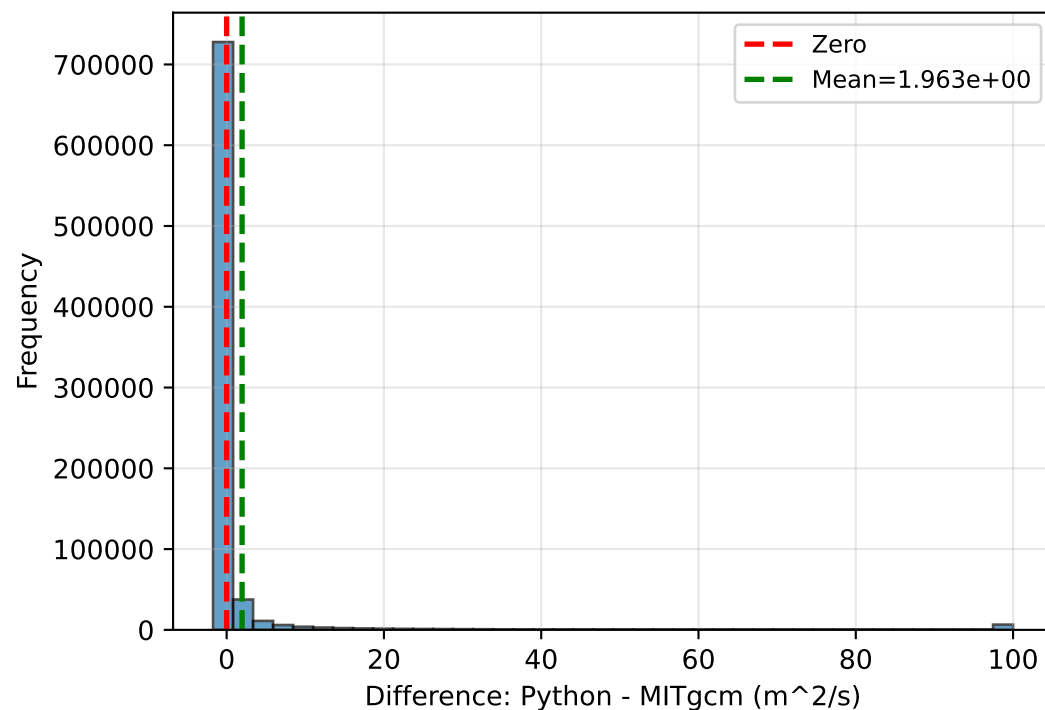
MITgcm Distribution



Scatter Plot (20000 of 813820 pts)



Difference Distribution



visc_az Statistics:

Median |diff|: 3.9072e-04
P95 |diff|: 4.9916e+00
Max |diff|: 1.0000e+02

Median rel: 3.6737e+04
P95 rel: 5.1923e+05
Max rel: 1.7473e+06

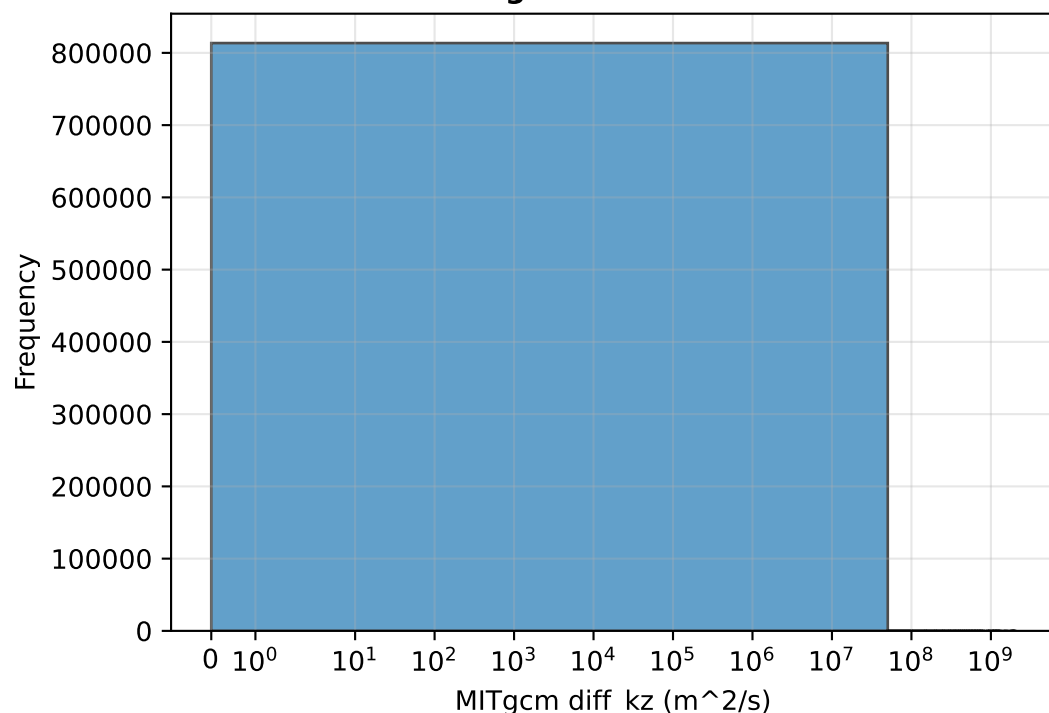
Fraction >1% rel: 62.5125%
Points compared: 813820
NaN-excluded: 107780

Assessment:

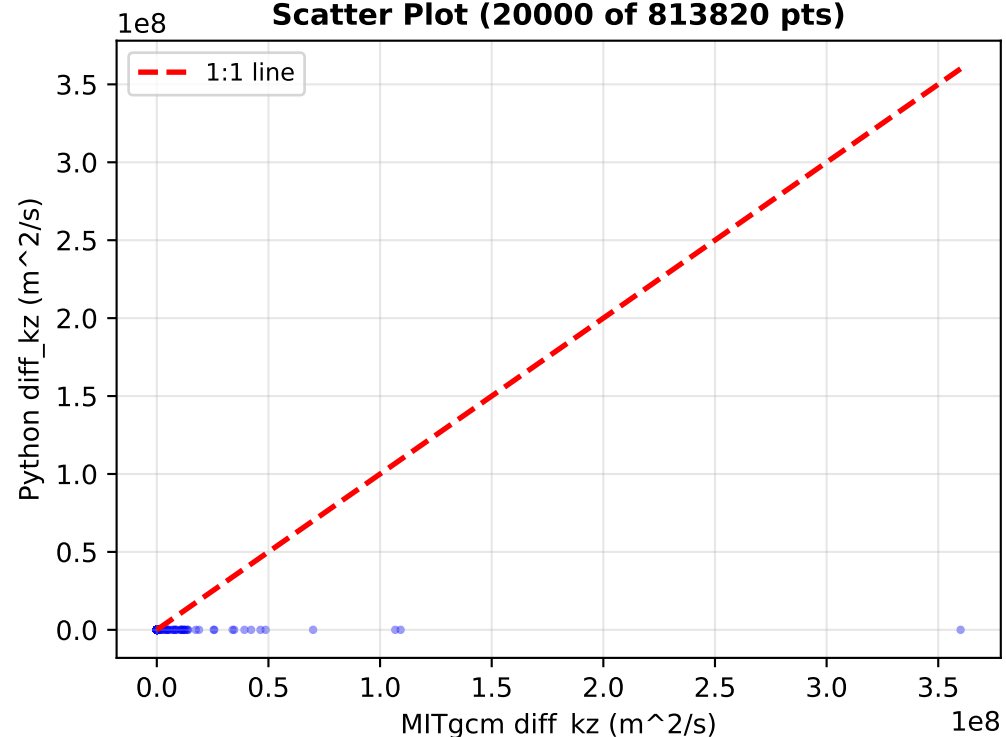
△ □ CHECK

diff_kz (Vertical eddy diffusivity (K_h)) -- MITgcm vs Python

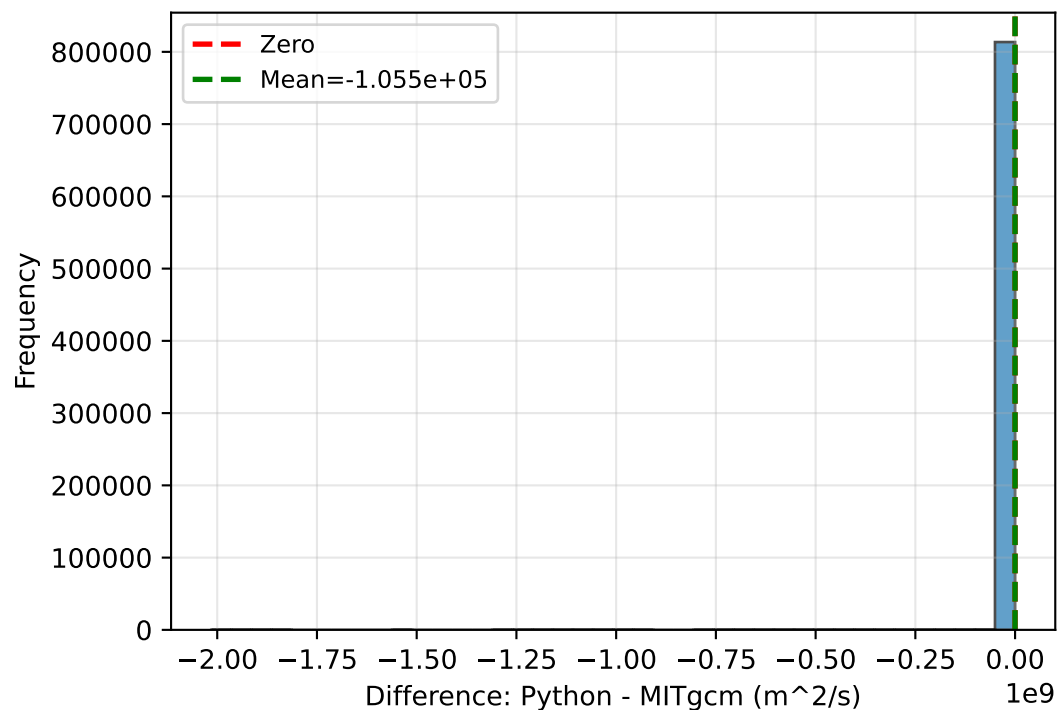
MITgcm Distribution



Scatter Plot (20000 of 813820 pts)



Difference Distribution



diff_kz Statistics:

Median |diff|: 7.4892e-01
P95 |diff|: 3.3267e+03
Max |diff|: 2.0149e+09

Median rel: 9.9695e-01
P95 rel: 9.9975e-01
Max rel: 1.0000e+00

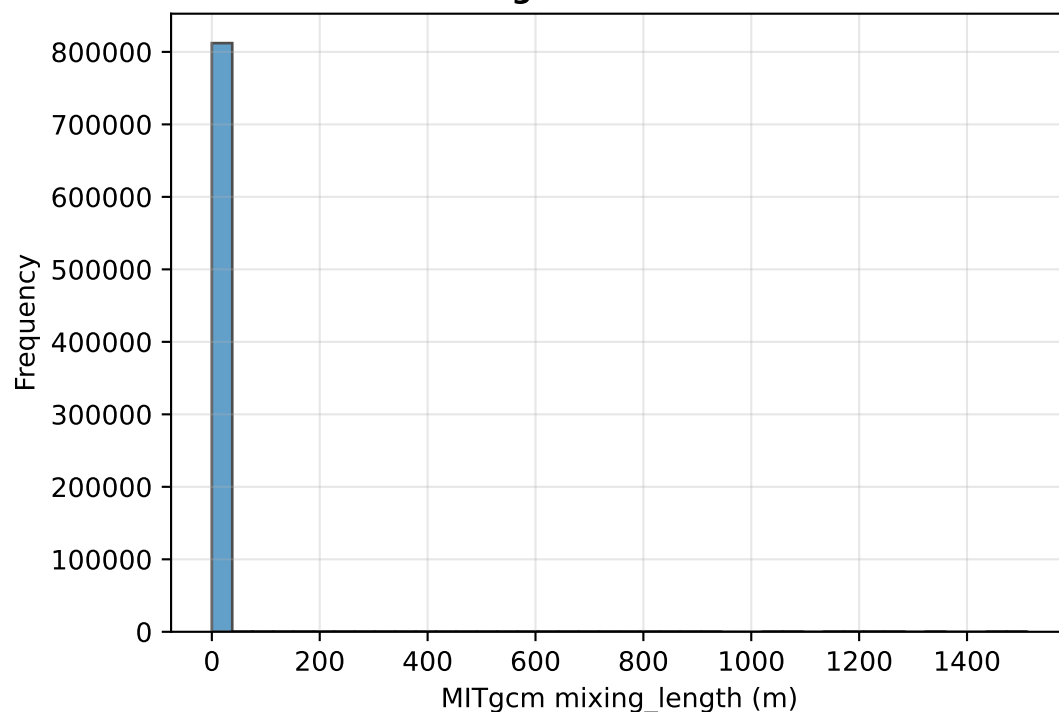
Fraction >1% rel: 62.7333%
Points compared: 813820
NaN-excluded: 107780

Assessment:

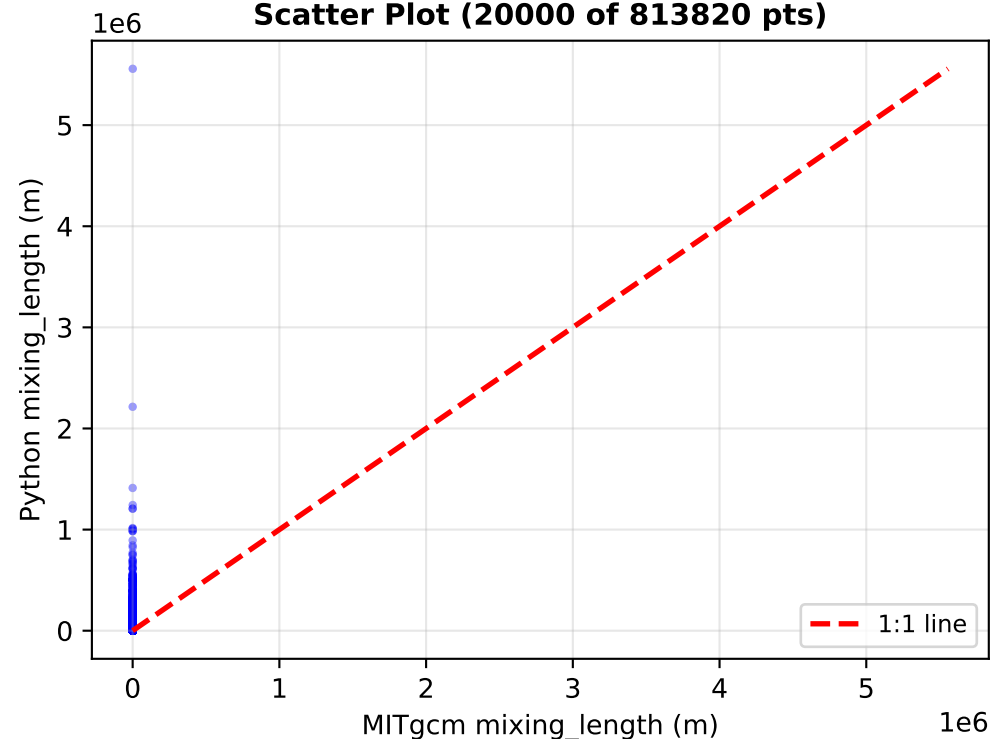
△ CHECK

mixing_length (GGL90 mixing length (L)) -- MITgcm vs Python

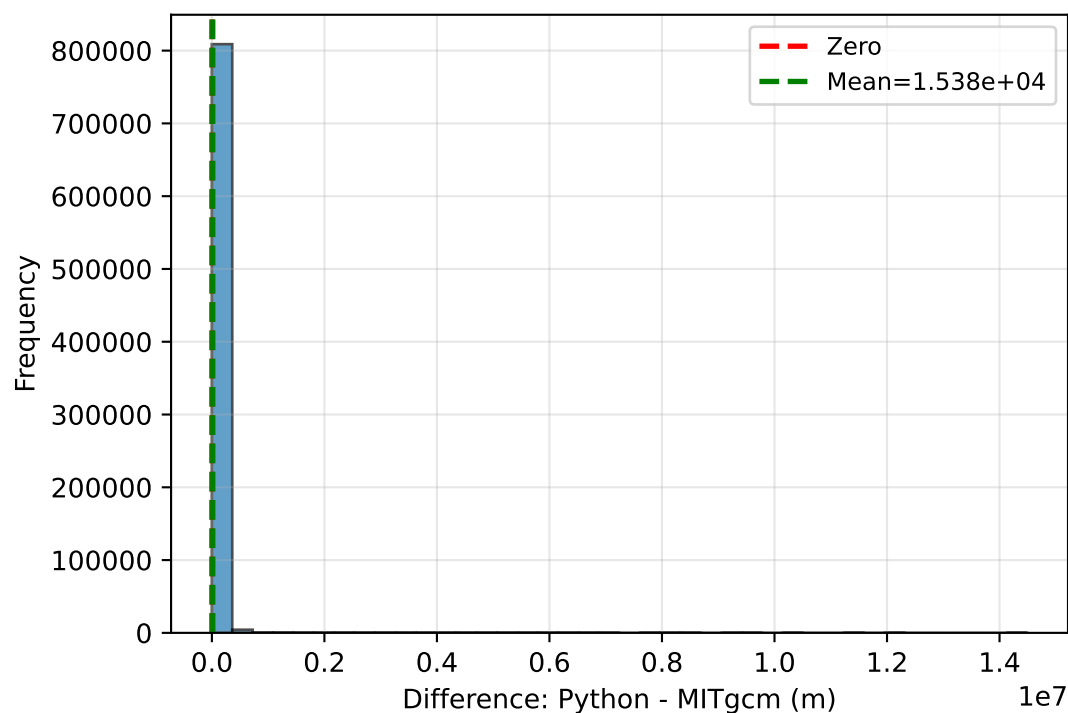
MITgcm Distribution



Scatter Plot (20000 of 813820 pts)



Difference Distribution



mixing_length Statistics:

Median |diff|: 5.1744e+02
P95 |diff|: 6.7815e+04
Max |diff|: 1.4492e+07

Median rel: 5.5952e+04
P95 rel: 5.5528e+05
Max rel: 1.7473e+06

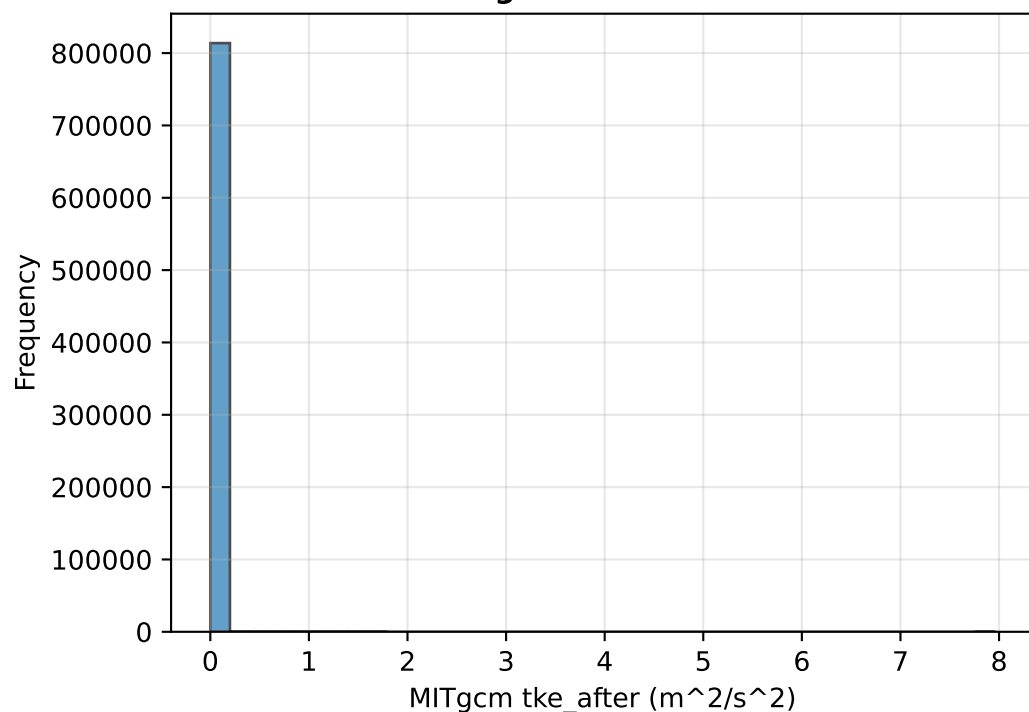
Fraction >1% rel: 62.7157%
Points compared: 813820
NaN-excluded: 107780

Assessment:

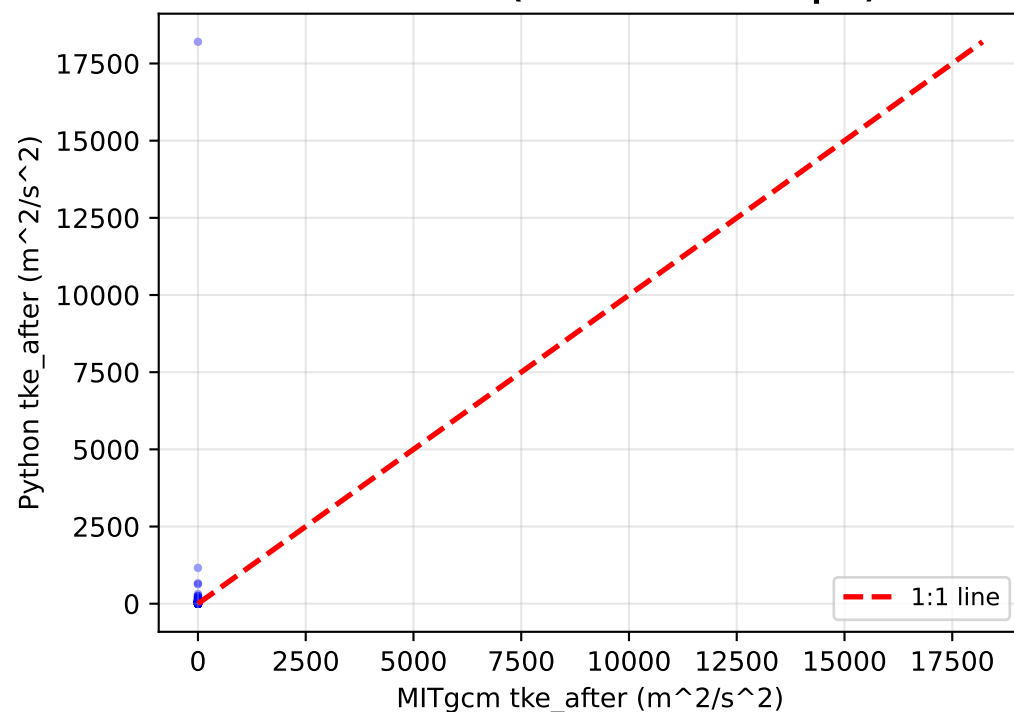
△ □ CHECK

tke_after (TKE at t+1 (single-step replay)) -- MITgcm vs Python

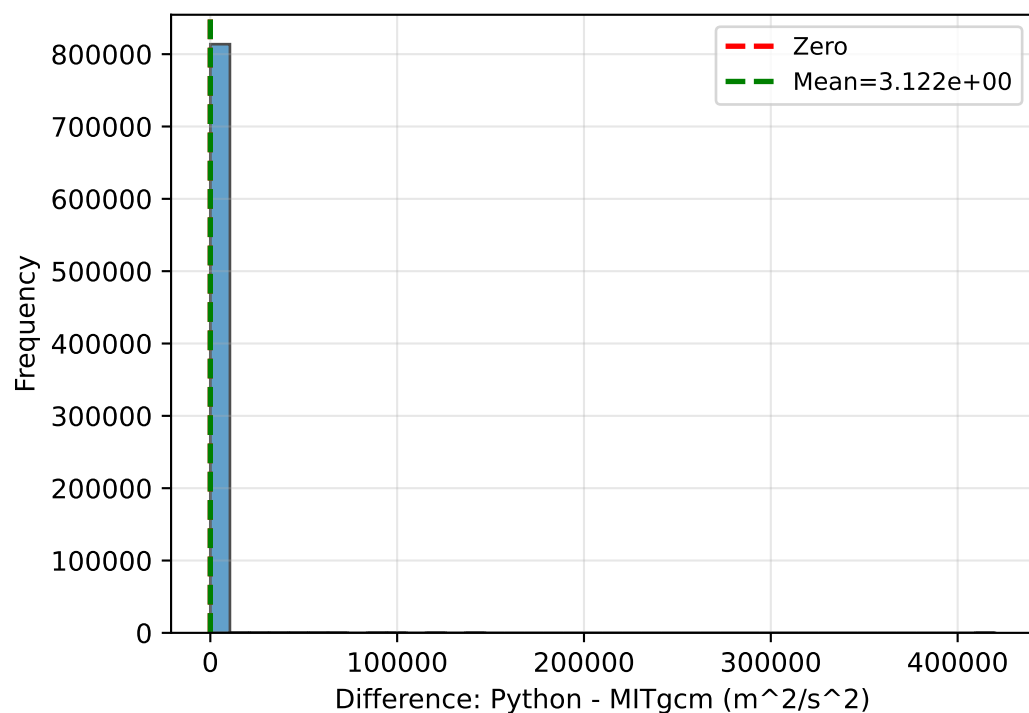
MITgcm Distribution



Scatter Plot (20000 of 813820 pts)



Difference Distribution



tke_after Statistics:

Median |diff|: 5.1499e-08
P95 |diff|: 5.3796e-02
Max |diff|: 4.1995e+05

Median rel: 3.0171e+01
P95 rel: 2.0207e+06
Max rel: 4.3265e+15

Fraction >1% rel: 64.7700%
Points compared: 813820
NaN-excluded: 107780

Assessment:

△ □ CHECK

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: Δ ☐ NEEDS REVIEW

COMPONENT RESULTS:

visc_az: Δ ☐ CHECK
Median rel error: 3673693.5323% Max |diff|: 1.0000e+02
Fraction >1% rel: 62.5125%

diff_kz: Δ ☐ CHECK
Median rel error: 99.6947% Max |diff|: 2.0149e+09
Fraction >1% rel: 62.7333%

mixing_length: Δ ☐ CHECK
Median rel error: 5595188.4970% Max |diff|: 1.4492e+07
Fraction >1% rel: 62.7157%

tke_after: Δ ☐ CHECK
Median rel error: 3017.1361% Max |diff|: 4.1995e+05
Fraction >1% rel: 64.7700%

PROVENANCE:

UUID Match: ☐ Yes
MITgcm UUID: 79a90418-e730-49e3-ac21-0cdae8b3015a
Python UUID: 79a90418-e730-49e3-ac21-0cdae8b3015a

ACCEPTANCE CRITERIA (mirrors generate_kpp_validation_report.py's own):

☐ EXCELLENT: median rel error < 0.1%
☐ GOOD: median rel error < 1.0%
 Δ ☐ CHECK: median rel error >= 1.0% -- may be a real, already-characterized gap (see closed_issues.md), not necessarily a new defect; this report states the number, not the cause.